

Week 1 - Discovering the Atmosphere 1: Worked Answers

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS:** selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY:** expected independent practice after the session.
- **EXTRA CHALLENGE:** optional problems for students who want additional depth or a harder application.

The categories are repeated here so the worked answers map directly onto the practice sheet. Use the solutions to check method, units and physical interpretation rather than only the final number. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

IN CLASS Question 1: Choose the observations you actually need

A forecaster has access to the following observing systems:

A	surface thermometer/hygrometer	E	weather radar
B	barometer	F	ceilometer
C	anemometer/wind vane	G	radiosonde
D	rain gauge	H	satellite imager

For each task below, choose the *single most useful* system from the list and justify your choice in one sentence. Where another observing system would provide genuinely useful complementary information, name *one additional system* and state what it adds.

1. Warn Manchester Airport about low cloud and poor visibility.
2. Determine whether a line of heavy showers is approaching Manchester during the next hour.
3. Measure the vertical temperature and humidity structure before a possible convective outbreak.
4. Establish whether surface pressure is falling rapidly as a depression approaches.

Worked answer

Reasonable choices are:

1. **Most useful: F (ceilometer)**, because it directly measures cloud-base height. **Also useful: A**, because temperature and humidity help diagnose near-surface saturation or fog. A dedicated visibility sensor would be even better, but it is not in the supplied list.
2. **Most useful: E (weather radar)**, because it shows the location, intensity and motion of precipitation. **Also useful: H (satellite)**, because it provides the wider cloud-system context.
3. **Most useful: G (radiosonde)**, because it directly measures the required vertical temperature and humidity structure. **Also useful: A**, because it provides representative surface conditions at launch time.
4. **Most useful: B (barometer)**, because it directly measures the pressure tendency. **Also useful: C**, because an approaching depression is commonly accompanied by systematic wind changes and strengthening flow.

The important point is to select measurements that constrain the particular atmospheric process rather than simply listing every available instrument.

YOU SHOULD TRY Question 2: Diagnose poor exposure at an automatic weather station

An automatic weather station has been installed with the following sensor placements.

Variable	Sensor placement
Air temperature	unshielded sensor fixed to a sunlit brick wall
Wind	anemometer 2 m above ground immediately behind a row of trees
Rainfall	tipping-bucket gauge beneath an overhanging branch
Pressure	calibrated electronic pressure sensor inside a ventilated instrument box

For each of the first three measurements, state the likely direction or nature of the bias and give one practical improvement. Why is the pressure measurement less affected by the external exposure problems listed here?

Worked answer

The temperature sensor is likely to read too warm during sunshine because it receives direct radiation and heat from the wall. It should be placed in a ventilated radiation shield over a representative surface and away from buildings.

The wind measurement will be unrepresentative because the trees slow, deflect and turbulently disturb the flow. The anemometer should be placed in a much more open exposure and at an appropriate height above surrounding obstacles.

The rain gauge is likely to undercatch because the branch shelters the gauge and can also cause splash/drip contamination. It should be moved to an unobstructed, level site with suitable exposure.

Pressure is much less sensitive to sunlight or local shelter because the sensor measures static air pressure within the instrument housing. It still requires good calibration and, for synoptic comparison, an appropriate height reduction to mean sea level.

IN CLASS Question 3: Why did fast communication transform forecasting?

A mid-latitude weather system is travelling eastwards at 15 m s^{-1} .

1. How far does it travel in 6 hours?
2. A set of observations takes 8 hours to reach a forecasting office. Would those observations still describe the same weather system at the location where they were made? Quantify your answer.
3. Use the calculation to explain why telegraphy was a major development for synoptic forecasting.

Worked answer

Six hours is 21600 s, so

$$d = vt = 15 \times 21600 = 3.24 \times 10^5 \text{ m} = 324 \text{ km}.$$

In 8 hours the system travels

$$15 \times (8 \times 3600) = 4.32 \times 10^5 \text{ m} = 432 \text{ km}.$$

Thus an observation delayed by 8 hours can describe a system that is already hundreds of kilometres downstream. Telegraphy allowed observations from many sites to be assembled quickly enough to construct a nearly simultaneous synoptic picture and issue warnings before the system arrived farther east.

IN CLASS Question 4: In situ or remote sensing? Active or passive?

Classify each observing system as *in situ* or *remote sensing*. For the remote-sensing systems, also state whether the instrument is mainly active or passive.

1. A radiosonde measuring temperature and humidity as it ascends.
2. A weather radar transmitting microwave pulses and receiving echoes from precipitation.
3. A ceilometer transmitting laser pulses to determine cloud-base height.
4. A satellite measuring naturally emitted infrared radiation from the Earth-atmosphere system.

Worked answer

1. Radiosonde: *in situ*; its sensors are immersed in the air being sampled.
2. Weather radar: active remote sensing; it transmits energy and measures the returned signal.
3. Ceilometer: active remote sensing; it transmits laser pulses and measures backscatter.
4. Infrared satellite imager/sounder: passive remote sensing; it detects naturally emitted radiation.

EXTRA CHALLENGE Question 5: Choose the satellite orbit for the job

For each task, choose either a *geostationary* or *polar-orbiting* satellite and give the single most important reason for your choice.

1. Follow the development of a rapidly intensifying Atlantic storm every 10–15 minutes.
2. Obtain detailed global temperature/humidity soundings, including high latitudes.
3. Monitor the movement of a convective cloud system across western Europe during one afternoon.

Then state one important limitation of geostationary observations at high latitude.

Worked answer

1. Geostationary: rapid repeat imaging of the same region is the key advantage.
2. Polar-orbiting: successive low-Earth-orbit passes provide near-global coverage and better access to high latitudes, often with higher spatial resolution and sounding capability.
3. Geostationary: frequent repeat images make it possible to track the evolution and motion of the cloud system continuously through the afternoon.

At high latitude a geostationary satellite views the surface at a very oblique angle, degrading spatial resolution and making retrievals more difficult.

YOU SHOULD TRY Question 6: Accuracy, precision and calibration

Two temperature sensors are tested in a laboratory bath held at exactly 20.0°C.

Sensor A reads 19.1, 19.2, 19.1, 19.2 and 19.1°C.

Sensor B reads 19.7, 20.4, 20.1, 19.8 and 20.0°C.

1. Calculate the mean reading from each sensor.
2. Which sensor is more precise?
3. Which sensor is more accurate overall?
4. What specific problem would calibration attempt to correct for Sensor A?

Worked answer

For Sensor A,

$$\bar{T}_A = \frac{19.1 + 19.2 + 19.1 + 19.2 + 19.1}{5} = 19.14^\circ\text{C}.$$

For Sensor B,

$$\bar{T}_B = \frac{19.7 + 20.4 + 20.1 + 19.8 + 20.0}{5} = 20.00^\circ\text{C}.$$

Sensor A is more precise because its repeated readings are tightly clustered. Sensor B is more accurate overall because its mean equals the reference value. Calibration of Sensor A would identify and correct its systematic negative offset of about 0.86°C ; calibration does not by itself remove random scatter.

IN CLASS Question 7: Can a sparse observing schedule miss a short-lived event?

A sharp wind shift associated with a shower line lasts for 20 minutes. A traditional synoptic station reports only at 00, 03, 06, 09 UTC, and so on, whereas a nearby automatic station stores one-minute values. Assume the 20-minute event is equally likely to begin at any time during a 3-hour interval between synoptic reports.

1. What is the probability that one of the 3-hourly synoptic observation times falls within the 20-minute event?
2. Hence, what is the probability that the event occurs entirely between two successive synoptic reports and is missed by that record?
3. The automatic station reports a wind shift between its 10:17 and 10:18 observations. To approximately what precision can the passage time be identified from this record?
4. Explain why temporal resolution should be chosen with reference to the timescale of the phenomenon being studied.

Worked answer

There are 180 minutes between successive 3-hourly reports. For an event lasting 20 minutes, the fraction of possible start times for which a scheduled report falls inside the event is approximately

$$P(\text{sampled}) = \frac{20}{180} = 0.111,$$

so the probability is about 11%.

The probability that it is missed is therefore approximately

$$1 - 0.111 = 0.889,$$

or 89%.

If the automatic record changes between 10:17 and 10:18, the passage can be located to within a one-minute interval (roughly ± 0.5 minute if a midpoint estimate is used).

A sampling interval much longer than the event timescale can alias or completely miss the event. A sampling interval much shorter than the event duration resolves both its timing and evolution.